Math **163** Name (PLEASE PRINT) Captain Yellowbeard

Exam **1C** **2024** Fall/Spring/**Summer I** /Summer II Score _____

Please fill in the following table putting in expected points corresponding to the parts of problems that you have completed and submitted and an expected total. Leave blank those parts that you have not completed or that do not exist. You will lose double the points that a part is worth if you submit incomplete, unproved, or unchecked answers, or if you do not follow the directions of the problem. Ideally you should know exactly what you have done correctly and how many points you should receive for parts and in total. I will fill in the received grade after I correct the exam. Not all questions have the same number of parts so that some columns may be redundant for particular problems.

										Expected	Received
	Part	Part	Part	Part	Part	Part	Part	Part	Part	Grade	Grade
	(i)	(ii)	(iii)	(iv)	(v)	(vi)	(vii)	(viii)	(ix)		
Number											
1a	2	2	2							6	
1b	3	3								6	
1c											
2a	3	3								6	
2b	6									6	
2c											
3a	1	1	1	1	1	1				6	
3b	6									6	
3c											
								Total		36	

Instructions:

Use these instructions for the remainder of the exam.

Your J-number

$$JN := (J_k \in 0..9 | k \in 1..8) = 00407205$$

Hence

$$J_1 = 0 \quad J_2 = 0 \quad J_3 = 4 \quad J_4 = 0 \quad J_5 = 7 \quad J_6 = 2 \quad J_7 = 0 \quad J_8 = 5$$

R Write **the elements of** every numerical set or multiset appearing in this exam **in increasing order.**

D $\mathcal{U} := 0..9 = \{0, 1, 2, 3, 4, 5, 6, 7, 8, 9\}$

D $SJ := St(J) := \left\{ J_k \in 0..9 \middle| k \in 1..8 \right\} = \{0, 2, 4, 5, 7\}$

$\nu(SJ) = 5$

D $Lst(J) := \left((SJ)_k \in 0..9 \middle| k \in 1..(\nu(SJ)) \right) = (0, 2, 4, 5, 7)$

D $\forall l \in 1..(\nu(SJ)) \quad n_l := \nu\left(\left\{ k \in 1..8 \middle| J_k = (SJ)_l \right\} \right)$

D $MSJ := MltSt(J) := \left\{ (SJ)_m : n_m \middle| m \in 1..(\nu(SJ)) \right\}$

$= \{0:4, 2:1, 4:1, 5:1, 7:1\}$

D $v(MSJ) = \sum_{m \in 1..(v(SJ))} n_m = 8$

D $s := mxm\{k \in 1..8 | J_k = 0\} = 7$

D $b := mnm\{J_k \in 0..9 | Evn(J_k)\} = 0$

D Define Y := Yes N := No Pf := Proof W := Witness

D $\forall x \in \mathbb{Z}$ $Evn(x)$ $:\Leftrightarrow$ x is even.

D $\forall x \in \mathbb{Z}$ $Odd(x)$ $:\Leftrightarrow$ x is odd.

D $\forall x \in \mathbb{N}$ $Prm(x)$ $:\Leftrightarrow$ x is prime.

R **Note that in a set an element may occur exactly once**. Therefore, there may not be any repeated elements in any set in the above. **So, if you do not have the correct sets, all your answers will be wrong.**

R If answers to different instances of some question on the exam contradict each other you will get a score of zero for every instance.

R **You will lose points for making any extraneous mark on the exam.**

R You may only submit complete answers to questions marked explicitly on the exam. You will lose points for unfinished work.

R Separate pieces of work for any part of any problem recorded on a blank page intended for the purpose must be indexed by the index of the problem and also by numbers starting at 0 that record the order in which the pieces were completed; such parts must be separated from each other by straight lines parallel to the edges of the paper drawn with a ruler. You will lose points for otherwise.

R Any sloppiness, untidiness, and any departure from proper format (as indicated in class) will lead to a score of 0.

6 (1a) Compute, $\forall n \in \mathbb{N}, \forall k \in 0..n$ the following, and express the result as a polynomial in n. **Show computations or substitutions into appropriate formulas on pages that follow, numbered as per instructions, and write the answers on this page. You must use your own numbers before computing.**

2 (i) $$C\left(C\left(v\left(St\left(J\right)\right), v\left(St\left(J\right)\right)\right), P\left(v\left(MltSt\left(J\right)\right), v\left(MltSt\left(J\right)\right)\right)\right)$$

$= \binom{1}{(8!)}$ does not make sense

2 (ii) $$C\left(P\left(v\left(MltSt\left(J\right)\right), v\left(St\left(J\right)\right)\right), P\left(v\left(St\left(J\right)\right), v\left(St\left(J\right)\right)\right)\right)$$

$$= \binom{\left(\frac{(8!)}{(3!)}\right)}{(5!)}$$

2 (iii) $$C\left(P\left(v\left(MltSt\left(J\right)\right), v\left(MltSt\left(J\right)\right)\right), P\left(v\left(St\left(J\right)\right), v\left(St\left(J\right)\right)\right)\right)$$

$$= \binom{(8!)}{(5!)}$$

1a.i.0

$$\frac{c(n,n) = 1}{c(v(St(J)), v(St(J))) = 1} \left\langle n \leftarrow v\left(St\left(J\right)\right) \right\rangle$$

1a.i.1

$$\frac{P(n,n) = n!}{P(v(MltSt(J)), v(MltSt(J))) = (v(MltSt(J)))!} \left\langle n \leftarrow v\left(MltSt\left(J\right)\right) \right\rangle$$

1a.i.2

$$\begin{aligned} & c\left(c\left(v(St(J)), v(St(J))\right), P\left(v(MltSt(J)), v(MltSt(J))\right)\right) \\ = \quad & c\left(1, \left((v(MltSt(J)))!\right)\right) \\ = \quad & c(1, 8!) \\ = \quad & \binom{1}{(8!)} \end{aligned}$$

1a.i.3

$$\frac{\forall n \in \mathbb{N} \quad \forall k \in 0..n \ \binom{n}{k}}{\binom{1}{(8!)} \text{ does not make sense } \because (8!) \notin 0..1}$$

1a.ii.0

$$\frac{P(n,m) = \frac{(n!)}{((n-m)!)}}{P(v(MltSt(J)), v(St(J))) = \frac{\left((v(MltSt(J)))!\right)}{\left(\left((v(MltSt(J))) - (v(St(J)))\right)!\right)}} \left\langle \begin{array}{l} n \leftarrow v(MltSt(J)) \\ m \leftarrow v(St(J)) \end{array} \right\rangle$$

1a.ii.1

$$\frac{P(n,n) = n!}{P(v(St(J)), v(St(J))) = (v(St(J)))!} \left\langle n \leftarrow v(St(J)) \right\rangle$$

1a.ii.2

$$\begin{aligned} & c\left(P\left(v(MltSt(J)), v(St(J))\right), P\left(v(St(J)), v(St(J))\right)\right) \\ = \quad & c\left(\left(\frac{\left((v(MltSt(J)))!\right)}{\left(\left((v(MltSt(J))) - (v(St(J)))\right)!\right)}\right), \left(v(St(J))\right)!\right) \\ = \quad & c\left(\left(\frac{(8!)}{(8-5)!}\right), (5!)\right) \\ = \quad & c\left(\left(\frac{(8!)}{(3!)}\right), (5!)\right) \\ = \quad & \left(\begin{pmatrix}(8!)\\(3!)\end{pmatrix}\right) \\ & \begin{pmatrix}(5!)\end{pmatrix} \end{aligned}$$

1a.iii.0

$$\frac{P(n,n) = n!}{P(v(MltSt(J)), v(MltSt(J))) = (v(MltSt(J)))!} \left\langle n \leftarrow v(MltSt(J)) \right\rangle$$

1a.iii.1

$$\frac{P(n,n) = n!}{P(v(St(J)), v(St(J))) = (v(St(J)))!} \left\langle n \leftarrow v(St(J)) \right\rangle$$

1a.iii.2

$$C\left(P(v(MltSt(J)), v(MltSt(J))), P(v(St(J)), v(St(J)))\right)$$

$$= C(((v(MltSt(J)))!), ((v(St(J)))!))$$

$$= C((8!), (5!))$$

$$= \begin{pmatrix} (8!) \\ (5!) \end{pmatrix}$$

6 (1b) **Provide complete reasoning by making a systematic, exhaustive list in the form of a table,** to generate, list, and count the following, **showing all work on pages that follow, numbered as per instructions, and writing only the answers on this page**:

R The symbol $\nmid$ means does not divide.

3 (i) D $F := \left\{ (x,y) \in \left(St\left(J \right) \right) \times \left(St\left(J \right) \right) \middle| x \nmid y \right\}$

compute: $F = \left\{ \begin{array}{c} (0,2), (0,4), (0,5), (0,7), (2,5), \\ (2,7), (4,2), (4,5), (4,7), (5,2), \\ (5,4), (5,7), (7,2), (7,4), (7,5) \end{array} \right\}$

compute: $\nu\left(F \right) = \underline{15}$

3 (ii) D $G := \left\{ (x,y) \in \left(St\left(J \right) \right) \times \left(St\left(J \right) \right) \middle| y \nmid x \right\}$

compute: $G = \left\{ \begin{array}{c} (2,0), (2,4), (2,5), (2,7), (4,0), \\ (4,5), (4,7), (5,0), (5,2), (5,4), \\ (5,7), (7,0), (7,2), (7,4), (7,5) \end{array} \right\}$

compute: $\nu\left(G \right) = \underline{15}$

1b.i

$$F := \{(x,y) \in (St(J)) \times (St(J)) \mid x \nmid y\}$$

(x,y)	0	2	4	5	7
0	(0,0)	(0,2)	(0,4)	(0,5)	(0,7)
2	(2,0)	(2,2)	(2,4)	(2,5)	(2,7)
4	(4,0)	(4,2)	(4,4)	(4,5)	(4,7)
5	(5,0)	(5,2)	(5,4)	(5,5)	(5,7)
7	(7,0)	(7,2)	(7,4)	(7,5)	(7,7)

$$F = \{(0,2), (0,4), (0,5), (0,7), (2,5), (2,7), (4,2), (4,5), (4,7), (5,2), (5,4), (5,7), (7,2), (7,4), (7,5)\}$$

1b.ii

$$G := \{(x,y) \in (St(J)) \times (St(J)) \mid y \nmid x\}$$

(x,y)	0	2	4	5	7
0	(0,0)	(0,2)	(0,4)	(0,5)	(0,7)
2	(2,0)	(2,2)	(2,4)	(2,5)	(2,7)
4	(4,0)	(4,2)	(4,4)	(4,5)	(4,7)
5	(5,0)	(5,2)	(5,4)	(5,5)	(5,7)
7	(7,0)	(7,2)	(7,4)	(7,5)	(7,7)

$$G = \{(2,0), (2,4), (2,5), (2,7), (4,0), (4,5), (4,7), (5,0), (5,2), (5,4), (5,7), (7,0), (7,2), (7,4), (7,5)\}$$

6 (2a) **Use a tree** (**which you should draw on pages that follow, numbered as per instructions**) to **generate**, **list**, and **count** the elements of the set of all **3-letter words** in the alphabet $\{s, b\}$ such that

My $\{s, b\} = \{ \quad 7, 0 \quad \}$

3 (i) order matters and repetition is allowed, and

$$Wrd\left(3, \{ \quad 7, 0 \quad \}, OM, RA\right) = \begin{Bmatrix} 000, 007, 070, 077, \\ 700, 707, 770, 777 \end{Bmatrix}$$

$$\nu\left(Wrd\left(3, \{ \ 7 \ , \ 0 \ \}, OM, RA\right)\right) = 8$$

3 (ii) order does **not** matter and repetition is allowed

$$Wrd\left(3, \{ \ 7 \ , \ 0 \ \}, \cancel{OM}, RA\right) = \begin{Bmatrix} 000, 007, \\ 077, 777 \end{Bmatrix}$$

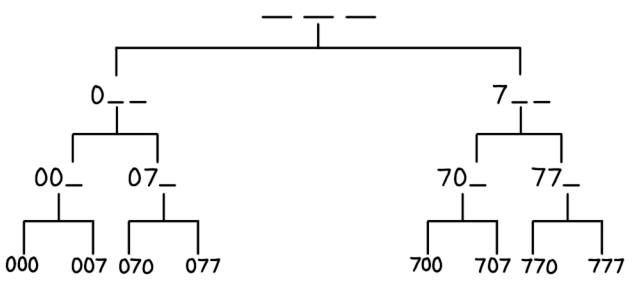
$$\nu\left(Wrd\left(3, \{ \ 7 \ , \ 0 \ \}, \cancel{OM}, RA\right)\right) = 4$$

2a

$$\{s,b\}$$
$$= (\{s,b\})\left\langle \begin{matrix} s \leftarrow 7 \\ b \leftarrow 0 \end{matrix} \right\rangle$$
$$= \{7,0\}$$

2a.i.0

Tree

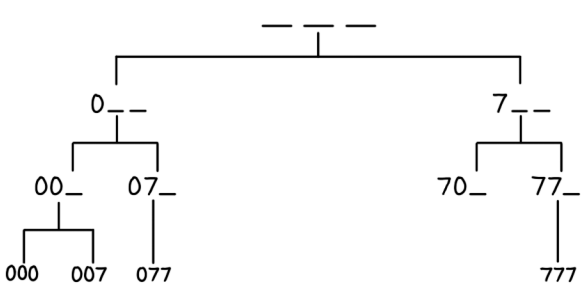


2a.i.1

$$\mathrm{Wrd}(3, \{7,0\}, \mathrm{OM}, \mathrm{RA}) = \{000, 007, 070, 077, 700, 707, 770, 777\}$$

2a.ii.0

Tree



2a.ii.1

$$\mathrm{Wrd}(3, \{7,0\}, \cancel{\mathrm{OM}}, \mathrm{RA}) = \{000, 007, 077, 777\}$$

6 (2b) Using a suitable tree (**that should be displayed on pages that follow, numbered as per instructions**) to **generate, count, and list** the elements of the set S of **distinct 4-digit** numbers that may be made by **using the elements of** $St\left(J \right)$ **as digits**:

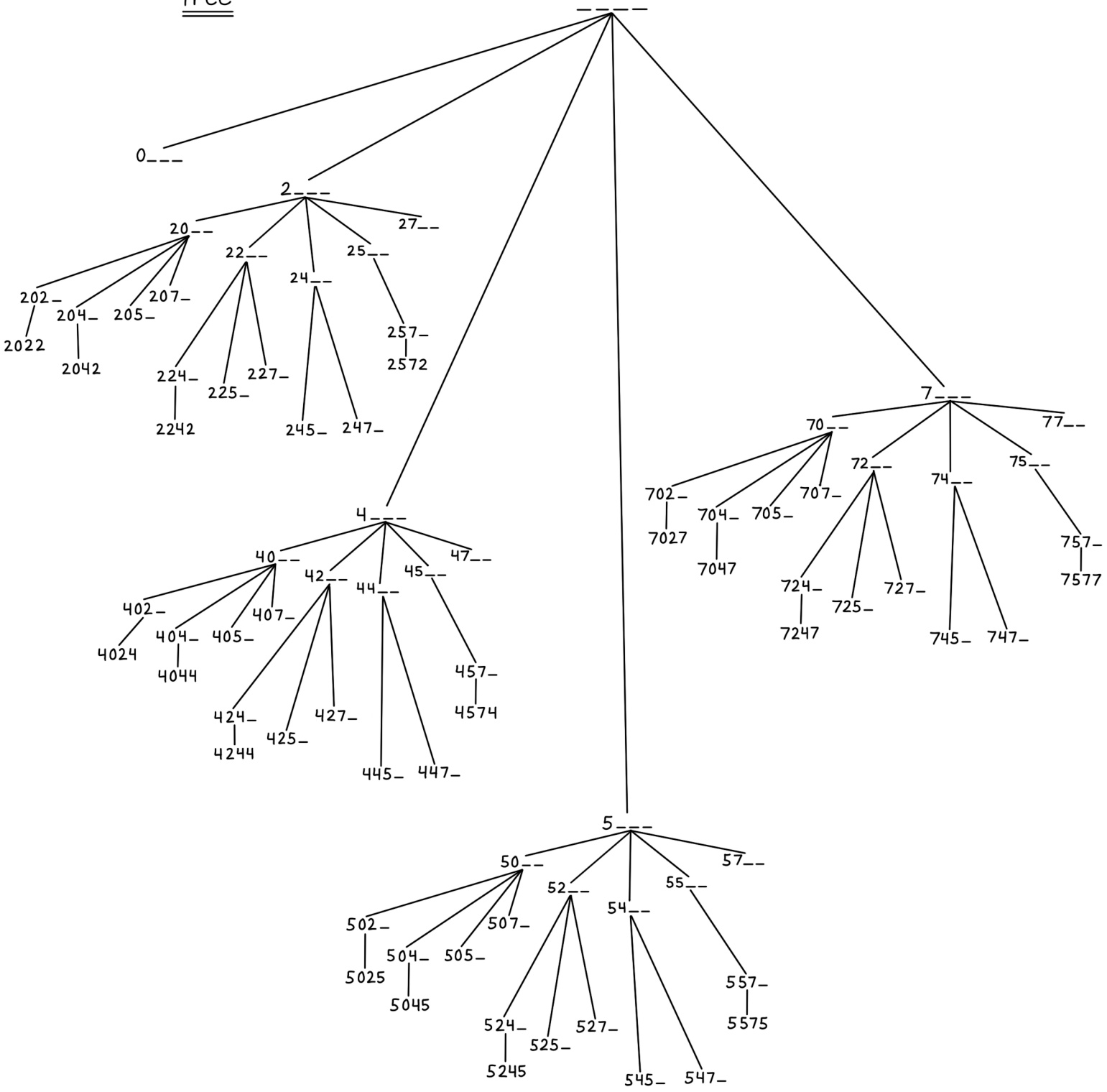
$$S = \left\{ \begin{array}{l} 2022, 2042, 2242, 2572, \quad 4024, 4044, 4244, 4574, \\ 5025, 5045, 5245, 5575, \quad 7027, 7047, 7247, 7577 \end{array} \right\}$$

such that:

- 0 cannot be used in the first place
- the same digit occupies the first and fourth place,
- the third digit is strictly larger than the second digit, and
- the sum of the digits is even.

E The 4-digit number 2352 since 2 occupies the first and fourth positions, and 5 is strictly larger than 3 and the sum of the digits is 12, an even number. The 4-digit number 2342 does not qualify as the sum of the digits is 11, not an even number. The number 2532 does not qualify as 3 is not strictly larger than 5. The number 2345 does not qualify as two distinct numbers occupy the first and fourth place.

Tree



$$S = \begin{Bmatrix} 2022, 2042, 2242, 2572, & 4024, 4044, 4244, 4574, \\ 5025, 5045, 5245, 5575, & 7027, 7047, 7247, 7577 \end{Bmatrix}$$

6 (3a) **Fill in the blanks appropriately** on this page and answer the following:

$$s = 7 \qquad b = 0 \qquad b + s = 7$$

1 (i) Compute, using the **binomial theorem**:

$$BnmlCfn\left(x^b y^s, (x+y)^{(b+s)}\right) = BnmlCfn\left(x^{(0)} y^{(7)}, (x+y)^{(7)}\right) = \binom{7}{7}$$

1 (ii) Compute, using the **binomial theorem**:

$$BnmlCfn\left(x^s y^b, (x+y)^{(s+b)}\right) = BnmlCfn\left(x^{(7)} y^{(0)}, (x+y)^{(7)}\right) = \binom{7}{0}$$

1 (iii) Compute, using the **multinomial theorem**:

$$MltnmlCfn\left(x^b y^s, (x+y)^{(b+s)}\right) = MltnmlCfn\left(x^{(0)} y^{(7)}, (x+y)^{(7)}\right) = \binom{7}{0\ 7}$$

1 (iv) Compute, using the **multinomial theorem**:

$$MltnmlCfn\left(x^s y^b, (x+y)^{(s+b)}\right) = MltnmlCfn\left(x^{(7)} y^{(0)}, (x+y)^{(7)}\right) = \binom{7}{7\ 0}$$

1 (v) The answers in (i), (ii), (iii) and (iv) are equal; record answer on this page and work on the facing page:

$$\textcircled{Y} \quad N \quad \textcircled{Pf} \quad W$$

1 (vi) The answer to (v) depend on the values of s and of b; record answer on this page and work on the facing page:

$$Y \quad \textcircled{N} \quad \textcircled{Pf} \quad W$$

3a

s=7	b=0	k=s	n−k=b
n−s=b			
n=b+s			
n=0+7			
n=7			

3a.i

$$\begin{aligned}
&\mathrm{BnmlCfn}(x^0 y^7, (x+y)^7) \\
&= \mathrm{BnmlCfn}(x^{7-7} y^7, (x+y)^7) \\
&= (\mathrm{BnmlCfn}(a^{n-k} b^k, (a+b)^n)) \left\langle \begin{matrix} n \leftarrow 7 \\ k \leftarrow 7 \\ a \leftarrow x \\ b \leftarrow y \end{matrix} \right\rangle \\
&= \left(\!\binom{n}{k}\!\right) \left\langle \begin{matrix} n \leftarrow 7 \\ k \leftarrow 7 \end{matrix} \right\rangle \\
&= \binom{7}{7}
\end{aligned}$$

3a.ii

$$\begin{aligned}
&\mathrm{BnmlCfn}(x^7 y^0, (x+y)^7) \\
&= \mathrm{BnmlCfn}(x^{7-0} y^0, (x+y)^7) \\
&= (\mathrm{BnmlCfn}(a^{n-k} b^k, (a+b)^n)) \left\langle \begin{matrix} n \leftarrow 7 \\ k \leftarrow 0 \\ a \leftarrow x \\ b \leftarrow y \end{matrix} \right\rangle \\
&= \left(\!\binom{n}{k}\!\right) \left\langle \begin{matrix} n \leftarrow 7 \\ k \leftarrow 0 \end{matrix} \right\rangle \\
&= \binom{7}{0}
\end{aligned}$$

3a.iii

$$\begin{aligned}
&\mathrm{MltnmlCfn}(x^0 y^7, (x+y)^7) \\
&= \left(\mathrm{MltnmlCfn}\left(\left(x_1^{n_1}\right)\left(x_2^{n_2}\right)\ldots\left(x_p^{n_p}\right), (x_1+x_2+\ldots+x_p)^n\right)\right) \left\langle \begin{matrix} p \leftarrow 2 \\ n \leftarrow 7 \\ n_1 \leftarrow 0 \\ n_2 \leftarrow 7 \\ x_1 \leftarrow x \\ x_2 \leftarrow y \end{matrix} \right\rangle \\
&= \left(\!\binom{n}{n_1\ n_2\ \ldots\ n_p}\!\right) \langle \rangle \\
&= \binom{7}{0\ \ 7}
\end{aligned}$$

3a.iv

$$\begin{aligned}
&\mathrm{MltnmlCfn}(x^7 y^0, (x+y)^7) \\
&= \left(\mathrm{MltnmlCfn}\left(\left(x_1^{n_1}\right)\left(x_2^{n_2}\right)\ldots\left(x_p^{n_p}\right), (x_1+x_2+\ldots+x_p)^n\right)\right) \left\langle \begin{matrix} p \leftarrow 2 \\ n \leftarrow 7 \\ n_1 \leftarrow 7 \\ n_2 \leftarrow 0 \\ x_1 \leftarrow x \\ x_2 \leftarrow y \end{matrix} \right\rangle \\
&= \left(\!\binom{n}{n_1\ n_2\ \ldots\ n_p}\!\right) \langle \rangle \\
&= \binom{7}{7\ \ 0}
\end{aligned}$$

3a.v.0

$$\begin{aligned}
&\binom{7}{7} \\
&= \frac{7!}{(7!)((7-7)!)} \\
&= \frac{7!}{(7!)(0!)}
\end{aligned}$$

3a.v.1

$$\begin{aligned}
&\binom{7}{0} \\
&= \frac{7!}{(0!)((7-0)!)} \\
&= \frac{7!}{(0!)(7!)}
\end{aligned}$$

3a.v.2

$$\begin{aligned}
&\binom{7}{0\ \ 7} \\
&= \binom{7}{7\ \ 0} \\
&= \frac{7!}{(0!)(7!)}
\end{aligned}$$

3a.v.3

$$\begin{aligned}
&\binom{7}{7} \\
&= \binom{7}{0} \\
&= \binom{7}{0\ \ 7} \\
&= \binom{7}{7\ \ 0}
\end{aligned}$$

3a.vi.0

$$\begin{aligned}
&\binom{b+s}{b} \\
&= \frac{(b+s)!}{(b!)(((b+s)-b)!)} \\
&= \frac{(b+s)!}{(b!)(s!)}
\end{aligned}$$

3a.vi.1

$$\begin{aligned}
&\binom{b+s}{s} \\
&= \frac{(b+s)!}{(s!)(((b+s)-s)!)} \\
&= \frac{(b+s)!}{(s!)(b!)}
\end{aligned}$$

3a.vi.2

$$\begin{aligned}
&\binom{b+s}{b\ \ s} \\
&= \binom{b+s}{s\ \ b} \\
&= \frac{(b+s)!}{(b!)(s!)}
\end{aligned}$$

3a.vi.3

$$\begin{aligned}
&\binom{b+s}{s} \\
&= \binom{b+s}{b} \\
&= \binom{b+s}{b\ \ s} \\
&= \binom{b+s}{s\ \ b}
\end{aligned}$$

6 (3b) Fill in the blank appropriately, and **show your work on pages that follow, numbered as per instructions**, to compute the value of, after **filling in the upper entry in the multinomial coefficient**:

$$\sum_{p+q=4} \binom{(4)}{0 \quad p \quad q}$$

$$= \qquad 2+(3!)+(2)\left(\frac{4!}{3!}\right)$$

3b.0

$$\frac{n = n_1 + n_2 + n_3}{n = 0 + p + q} \left\langle \begin{matrix} n_1 \leftarrow 0 \\ n_2 \leftarrow p \\ n_3 \leftarrow q \end{matrix} \right\rangle$$

$$\frac{n = 0 + (p+q)}{n = 0 + 4} \left\langle (p+q) \leftarrow 4 \right\rangle$$

$$n = 4$$

3b.1

$$\begin{aligned} 4 &= 0 + 0 + 4 \\ &= 0 + 1 + 3 \\ &= 0 + 2 + 2 \\ &= 0 + 3 + 1 \\ &= 0 + 4 + 0 \end{aligned}$$

3b.2

$$\sum_{p+q=4} \binom{4}{0\ p\ q}$$

$$= \begin{vmatrix} \binom{4}{0\ 0\ 4} \\ + \binom{4}{0\ 1\ 3} \\ + \binom{4}{0\ 2\ 2} \\ + \binom{4}{0\ 3\ 1} \\ + \binom{4}{0\ 4\ 0} \end{vmatrix} = \begin{vmatrix} 1 \\ + \frac{4!}{3!} \\ + 3! \\ + \frac{4!}{3!} \\ + 1 \end{vmatrix} = 2 + (3!) + (2)\left(\frac{4!}{3!}\right)$$

3b.3

$$\binom{4}{0\ 0\ 4}$$

$$= \binom{4}{0\ 4\ 0}$$

$$= \frac{4!}{(0!)(0!)(4!)}$$

$$= \frac{4!}{(1)(1)(4!)}$$

$$= \frac{4!}{4!}$$

$$= 1$$

3b.4

$$\binom{4}{0\ 1\ 3}$$

$$= \binom{4}{0\ 3\ 1}$$

$$= \frac{4!}{(0!)(3!)(1!)}$$

$$= \frac{4!}{(1)(3!)(1)}$$

$$= \frac{4!}{3!}$$

3b.5

$$\binom{4}{0\ 2\ 2}$$

$$= \frac{4!}{(0!)(2!)(2!)}$$

$$= \frac{4!}{(1)(2)(2)}$$

$$= \frac{\cancel{(4)}(3)(2)(1)}{\cancel{4}}$$

$$= 3!$$